

Four Core Pipelined Implied Volatility & Greeks Engine (Artix 7 200T @ 100 MHz)

**AXI4-Stream Ingress
(256-bit S, K, T, r, C, TID)**

**Stage 1: Scale Invariance
& Input Ingress**

AXI4-Stream 256-bit Ingress
 $S_tilde = S / K$, $K_tilde = 1.0$
 $C_tilde = C / K$
Q8.24 Overflow Protection

Zero-BRAM Context Store

64 x 32-bit Distributed LUTRAM
SRL32 Shift Register Pipeline
Zero RAMB36/18 Consumed
MAC/DMA BRAM Preserved

**Stage 2: Analytical Seeder
(Brenner-Subrahmanyam)**

$\sigma_0 = \sqrt{2\pi/T} * (C / S)$
Shared $\sqrt{T}$ Generator
Compact 32-bit Restoring Div
Eliminates 2nd $\sqrt{r}$ core

Stage 3: Newton-Raphson Engine

Padé Log: $\ln(S/K) \sim 2*(S-K)/(S+K)$
Hyperbolic CORDIC $\exp(x)$
Horner 5th-order Poly CDF $N(d)$
 $\Delta = (C_model - C_mkt) / Vega$

**Stage 4: Multi-Core Arb
& 128-bit Egress Bus**

Round-Robin Core Arbiter
[31:0] σ (Implied Vol)
[63:32] Δ (First Order)
[95:64] Vega (Vol Sens)
[121:96] Gamma (Second Order)
[127:122] TID Transaction ID